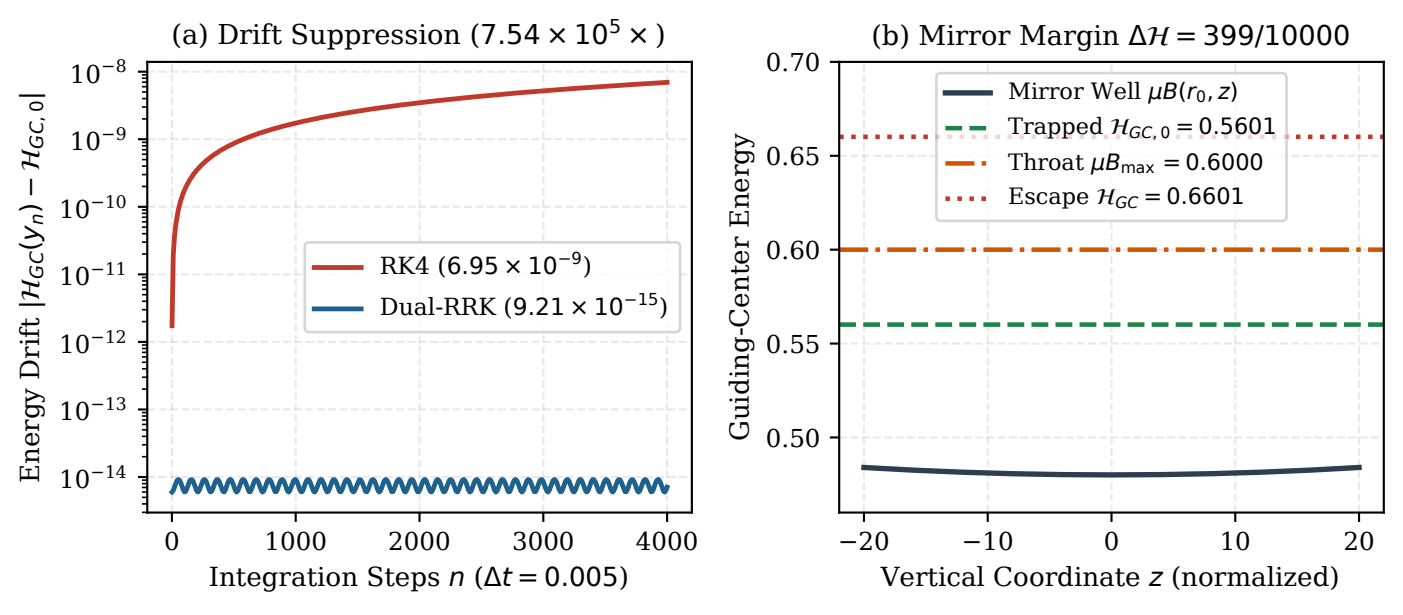
(a) Drift Suppression ($7.54 \times 10^5 \times$)
Energy Drift $|\mathcal{H}_{GC}(y_n) - \mathcal{H}_{GC,0}|$
10^{-8}
10^{-9}
10^{-10}
10^{-11}
10^{-12}
10^{-13}
10^{-14}
RK4 (6.95×10^{-9})
Dual-RRK (9.21×10^{-15})
0
1000
2000
3000
4000
Integration Steps n ($\Delta t = 0.005$)
(b) Mirror Margin $\Delta\mathcal{H} = 399/10000$
Guiding-Center Energy
0.70
0.65
0.60
0.55
0.50
Mirror Well $\mu B(r_0, z)$
Trapped $\mathcal{H}_{GC,0} = 0.5601$
Throat $\mu B_{\max} = 0.6000$
Escape $\mathcal{H}_{GC} = 0.6601$
−20
−10
0
10
20
Vertical Coordinate z (normalized)